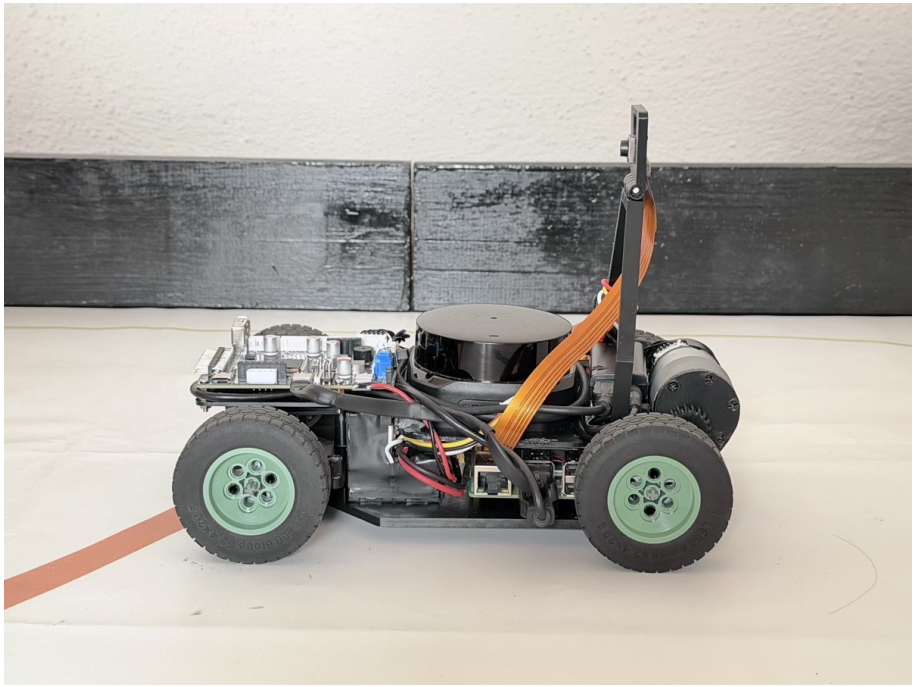
ENGINEERING JOURNAL

# Team CSS

## WRO Future Engineers 2026

### International Final edition · Puerto Rico, December 2026

Querétaro, Mexico



*The international competition vehicle, **"The Maker"** (design v7)*

| | |
|---|---|
| Vehicle | "The Maker", design v7 (international). Design v6 placed 2nd at the 2026 Mexico National Final |
| Team members | Christian Gael Centeno Velez (Captain) · Sebastian Esquivel Mondragon |
| Coach | Manuel Alejandro Cardoso Duarte |
| Dimensions / Mass | 238.5 × 157.7 × 180 mm · 1.087 kg |
| Drive / Steering | Rear-wheel drive with differential, JGB37-520 30:1 encoder motor · Front Ackermann, HX-35H bus servo |
| Compute | Raspberry Pi 5 + Hailo-8 · Hiwonder ROS Robot Control Board (STM32F407) |
| Sensors | Slamtec RPLidar S2L · Raspberry Pi Camera Module 3 Wide · IMU on the Hiwonder board |
| Power | 3 × Panasonic NCR18650B in series (3S, 10.8 V nominal) with 3S BMS |
| Repository | github.com/alex309-duarte/WRO_Future_Engineers_Queretaro_CSS (release v2.0.0) |

# How to read this journal

This is the **international edition** of our engineering journal. The national edition (September 2026, design v6) is preserved in the repository as release **v1.0.0**. This edition keeps the full history, because the reasons behind the national design are still the reasons behind most of design v7, and adds what we changed for the international final, why, and how we tested it.

Like the national edition, it is organised around the five evaluation criteria in Appendix C of the 2026 Future Engineers rules:

| Rubric criterion | Where it is addressed |
|---|---|
| 1. Mobility and Mechanical Design | Section 2 - steering, drivetrain, motor sizing (v6 and v7), differential, chassis, design evolution v1-v7 |
| 2. Power and Sensor Architecture | Section 3 - power budgets (v6 and v7), failures, LiDAR decisions in both directions, placement, calibration |
| 3. Software Architecture and Obstacle Strategy | Section 4 - threading, hardware abstraction layer, perception, control, strategies, vision model, emergency stop |
| 4. Systems Thinking and Engineering Decisions | Section 5 - tradeoff register, risk register, subsystem coupling |
| 5. Reproducibility and GitHub Quality | Sections 6-8 - testing workflow, dated logbook, repository map, interfaces, versions |

### Design numbers

We number every vehicle we have built since 2025, so **v1-v7 are vehicle designs**, not document versions: v1 and v2 are the 2025 robots, v3-v6 the 2026 national iterations, and **v7** is the international vehicle. Repository releases use their own numbers: release v1.0.0 is design v6 and release v2.0.0 is design v7.

### What changed since the national edition

| Subsystem | Design v6 (national) | Design v7 (international) | Section |
|---|---|---|---|
| Motor controller | LEGO SPIKE Prime Hub | Hiwonder ROS Robot Control Board (STM32F407) | 4.9 |
| Drive motor | LEGO Large Angular Motor, 175 rpm | JGB37-520 30:1, 320 rpm, Hall encoder | 2.3 |
| Rear differential | v6 differential | New differential, same 20:28 ratio | 2.2 |
| Steering actuator | LEGO Small Angular Motor | HX-35H serial bus servo | 2.1 |
| Heading | SPIKE Hub IMU | Hiwonder board IMU (20 Hz) | 4.4 |
| Battery | 4 × NCR18650B + Geekworm X1203 | 3 × NCR18650B in series (3S) + BMS | 3.1 |
| LiDAR | Oradar MS200k | RPLidar S2L (brought back) | 3.4 |
| AI accelerator / model | Hailo-10H, 501-image model | Hailo-8, 23,938-image model | 3.6, 4.8 |
| Safety | No emergency stop (open risk) | Signal-based emergency stop | 4.10 |
| Chassis | Custom, SolidWorks | Modified for the new parts and lowered | 2.4 |

**Every constant quoted in this journal is the value present in the source code in our public repository**, not a design target.

# 1. Design philosophy and season constraints

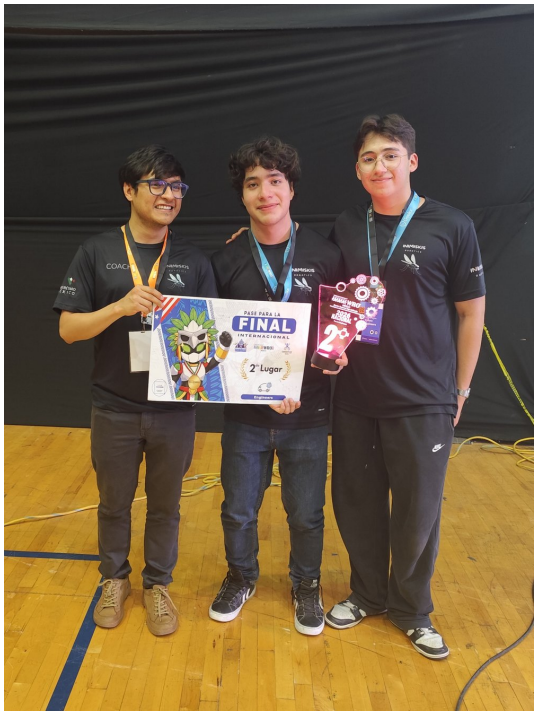
We did not start this season from a blank page. We finished the 2025 season with a working Future Engineers vehicle that placed 2nd nationally, and our starting decision was whether to iterate on it or rebuild. We chose to keep the concept - Ackermann-steered, LiDAR-led, LEGO SPIKE actuation - and rebuild everything else, because our post-season review identified three problems that were structural rather than tuning issues:

- **Latency.** The 2025 codebase was Python. Adding a higher-rate camera and LiDAR made the control loop the bottleneck, not the sensors.
- **Mass and packaging.** The 2025 vehicle came in barely under the 1.5 kg limit, with a high centre of gravity and loosely packaged electronics.
- **Mechanical inconsistency.** The custom double-Ackermann steering we built in 2025 gave a tighter turning radius but did not behave the same way twice.

Those findings became the season's design goals: **cut latency, cut mass, and trade peak capability for repeatability wherever the two conflict.**

## 1.1 Goals for the international final

After the national final (2nd place with design v6) we kept the same principle and added one goal: **more speed without losing repeatability.** The LEGO drivetrain reached only 0.365 m/s (Section 2.3). For the international final we replaced the LEGO SPIKE actuation with a faster encoder motor, a bus servo and a dedicated motor-control board, and we required that the mission logic that placed 2nd nationally be reused rather than rewritten (Section 4.9), because there were less than eight weeks between the national final and the repository deadline for the international final (10 November 2026).




*Team CSS with coach Manuel Alejandro Cardoso Duarte after the 2026 Mexico National Final: 2nd place and qualification for the International Final.*

## 1.2 Constraints we designed against

| Constraint | Source | Limit | v6 value | v7 value | How it shaped the design |
|---|---|---|---|---|---|
| Maximum dimensions | Rule 11.1 | 300 × 200 × 300 mm | 240 × 160 × 250 mm | 238.5 × 157.7 × 180 mm | Dense packaging; Hailo-8 chosen partly to reduce size |
| Maximum mass | Rule 11.2 | 1.5 kg | 1.006 kg (67 %) | 1.087 kg (72 %) | Mass freed in v7 allowed the S2L back |
| One driving axle, one steering actuator | Rule 11.3 | - | RWD + front Ackermann | RWD (differential) + front Ackermann | Ruled out AWD and two drive motors |
| No wireless during rounds | Rule 11.10 | - | Ethernet only | Ethernet only | See Section 5.2 |
| Track wall height | Game field | 100 mm | LiDAR at 60-80 mm | Chassis lowered for the LiDAR | Hard constraint - Section 3.5 |
| Round time limit | Rules 9.1 / 9.2 | 3 minutes | - | - | Minimum average speed for 3 laps |
| Time to the international repository deadline | Calendar | < 8 weeks (19 Sep - 10 Nov) | - | - | Reuse the national mission code (4.9) |
| Budget | Team | - | $33,018.73 MXN | Reused Pi 5, camera, tyres, cells and button from v6 | No spare vehicle; parts carried over where possible |

The mass figure deserves emphasis. Going from barely under 1.5 kg to 1.006 kg in v6 was the root cause of half the changes in Section 2. In v7 the same margin paid off again: removing the SPIKE Hub, the X1203 board and one battery cell left room for the heavier RPLidar S2L (185 g) without approaching the limit.

The parking bay is defined as 1.5× our own vehicle length, so making the robot shorter makes the bay proportionally shorter too. Shrinking the chassis does not make parking easier; it holds the difficulty constant while shrinking our margin for error in absolute millimetres.

# 2. Mobility and mechanical design

## 2.1 Steering: why we went backwards on purpose

In 2025 we designed and built a custom **double-Ackermann** linkage - both axles steering - to get a tighter turning radius. It worked, and on a good run it was better than a conventional layout. The problem was that it was not consistent: the additional linkage introduced play, and the steering did not return to the same neutral point reliably across runs.

For 2026 we replaced it with a **regular Ackermann** linkage steering the front axle only, reaching a maximum steer angle of **45.44°** in design v6. The design v7 linkage, now driven by a bus servo, reaches **45.9°** in the CAD, so the turning radius is the same as in v6; in software the steering is limited to ±35°.

> **We chose regular Ackermann instead of double-Ackermann because** the double-Ackermann's advantage was a smaller turning radius, but our failure mode was never "the corner was too tight to make" - it was "the robot did not end the corner pointing where we expected." The regular Ackermann completes every turn the WRO track requires with margin, and it removes a mechanical error source from every corner of a 12-corner run.

The Ackermann condition - inner and outer wheels tracing concentric arcs about a common centre - keeps the front tyres rolling instead of scrubbing. Scrubbing costs traction mid-corner, and it makes the effective wheelbase behave differently at different steer angles, which is exactly the non-repeatability we were trying to design out.

### *Design v6 actuator*

Steering was driven by a LEGO SPIKE Small Angular Motor (45607), replacing the Medium Angular Motor used previously. This was a consequence of the mass reduction: with a lighter chassis the required steering torque dropped enough that the smaller motor became sufficient.

### *Design v7 actuator: HX-35H serial bus servo*

With the SPIKE Hub removed (Section 4.9), the steering needed an actuator the Hiwonder board can command. We use a **Hiwonder HX-35H** serial bus servo on the same Ackermann linkage.

| HX-35H (Hiwonder) | Value | Why it matters for us |
|---|---|---|
| Voltage | 9-12.6 V | Runs directly from the 3S battery, no extra regulator |
| Stall torque | 35 kg·cm at 11.1 V | Large margin over the linkage load |
| Speed | 0.18 s/60° at 11.1 V | Fast steering response at battery voltage |
| Range / command | 0-240°, 0-1000 (0.24° per unit) | Matches kGradosPorUnidadServo = 0.24 in the code |
| Feedback | Position, temperature, voltage | Start-up check of the servo position |
| Stall current / mass | 3 A / 52 g | Counted in the peak-current analysis (3.1) |

> **We chose a serial bus servo instead of a PWM hobby servo because** the bus servo closes its own position loop, reports its position, and shares the serial bus with the board, so the board can both command and verify the steering. It runs at battery voltage, and its torque margin means the steering reaches its commanded angle at the same speed regardless of tyre grip on the mat.

Safety in software: every steering command passes through one function (*set_steering*) that clamps to **±35°** from centre (35 / 0.24 = 145 servo units each side, positions 605-895), the program refuses to move the servo at start-up if it is outside the allowed range, and the emergency stop releases the servo torque so it cannot push the linkage against an end stop.

## 2.2 Drivetrain: AWD to RWD, and a new differential

The 2025 vehicle was all-wheel drive. The 2026 vehicles are rear-wheel drive.

> **We chose RWD instead of AWD because** AWD required a front differential and driveshaft to carry torque past the steering axle. That hardware cost mass, volume in exactly the region where the Ackermann linkage lives, and one more set of mechanical tolerances. On a flat, high-grip competition mat the traction AWD buys is not the limiting factor.

### *Design v7: rear differential*

Design v7 uses a **new rear differential**: the LEGO 65414 differential housing with the 65413 28-tooth bevel gear in a custom mount (CAD: Diferencial.SLDASM). The gear ratio stays 20:28, so the speed and torque calculation of Section 2.3 and the distance constants of the missions keep their meaning.

> **We replaced the v6 differential instead of adapting it because** its shape and size did not fit the modified chassis and the new JGB37 motor well; the new one fits the structure and the motor mount better. We keep a differential at all because in a corner the outer rear wheel travels farther than the inner one: without it one wheel must slip, which wastes traction, wears the tyres and makes the encoder-based distance less accurate.

## 2.3 Motor sizing: the calculation we did before buying anything

### *Design v6: LEGO Large Angular Motor*

We sized the traction motor against a dynamic torque budget rather than picking a motor and hoping. Target: reach cruising speed in t = 0.5 s, implying a = 0.73 m/s². Inputs: m = 1.006 kg, r = 0.0312 m, $C_{rr}$ = 0.03, gear reduction 20:28 (R = 1.4), efficiency 0.85.

- Acceleration force $F_{acc}$ = m·a = 0.7344 N; rolling resistance $F_{rr}$ = $C_{rr}$·m·g = 0.2961 N; total 1.0305 N
- Wheel torque $T_w$ = 0.03215 N·m; motor torque $T_m$ = $T_w$/(η·R) = 0.02702 N·m (2.70 N·cm); with a 1.25× margin 3.38 N·cm

| Metric (LEGO 45602) | Value | Interpretation |
|---|---|---|
| Load vs max-efficiency torque (8.00 N·cm) | 33.8 % | Motor runs well inside its efficient band |
| Load vs stall torque (25.00 N·cm) | 10.8 % | No realistic stall risk |
| Loaded top speed | 0.365 m/s | Reached in 0.5 s at the design acceleration |
| Torque reserve | 66.2 % | Absorbs perturbations without dropping speed |

### *Design v7: Hiwonder JGB37-520, 30:1*

For the international final we wanted more speed while keeping the same principle: a motor that works far below its limit holds its speed when the load changes. We chose the Hiwonder JGB37-520R30-12V Hall-encoder gear motor.

> **We chose the JGB37 instead of keeping the LEGO motor because** the SPIKE motor was too slow for the level we expect at the international final, where every team is the best of its country. We chose this motor in particular because we already had a similar Hiwonder motor and were familiar with it, and its speed and torque cover our needs with margin (calculation below).

| JGB37-520R30-12V (Hiwonder) | Value |
|---|---|
| Rated voltage / gear ratio | 12 V / 30:1 |
| No-load speed | 320 rpm |
| Rated speed / torque / current | 280 rpm / 1.2 kg·cm / 0.36 A |
| Stall torque / current | 5.8 kg·cm / 3.2 A |
| Encoder | Hall, AB quadrature, 11 pulses per motor revolution → 11 × 30 × 4 = 1320 counts per output revolution (matches the 1320 ticks/rev we measured) |
| Mass | 152 g |

Same method as v6, with the measured v7 mass **m = 1.087 kg**, a = 1.3 m/s², r = 0.0312 m (LEGO 62.4 × 20 S tyres), the same 20:28 gear reduction as v6 (R = 1.4), η = 0.85, and the 3S nominal voltage of 10.8 V:

- $F_{acc}$ = 1.087 × 1.3 = 1.413 N; $F_{rr}$ = 0.03 × 1.087 × 9.81 = 0.320 N; $T_w$ = 1.733 × 0.0312 = 0.0541 N·m

- $T_m$ = 0.0541 / (0.85 × 1.4) = 0.0454 N·m (0.46 kg·cm); with the 1.25× margin 0.0568 N·m (0.58 kg·cm)
- No-load speed at 10.8 V: 320 × 10.8/12 = 288 rpm; loaded: 288 × (1 − 0.0454/0.569) = 265 rpm; wheel: 265 / 1.4 = 189 rpm

| Metric (JGB37 30:1) | Value | Interpretation |
|---|---|---|
| Load vs rated torque (1.2 kg·cm) | 39 % (48 % with margin) | Inside the continuous rating |
| Load vs stall torque (5.8 kg·cm) | 8 % | No stall risk in normal driving |
| Loaded top speed at 10.8 V | 0.62 m/s | 1.7× design v6 |
| Time to top speed | 0.48 s | Meets the 0.5 s target |
| Same check at the 1.5 kg rule limit | 53 % of rated torque, 0.60 m/s | The motor still has margin if mass is added |

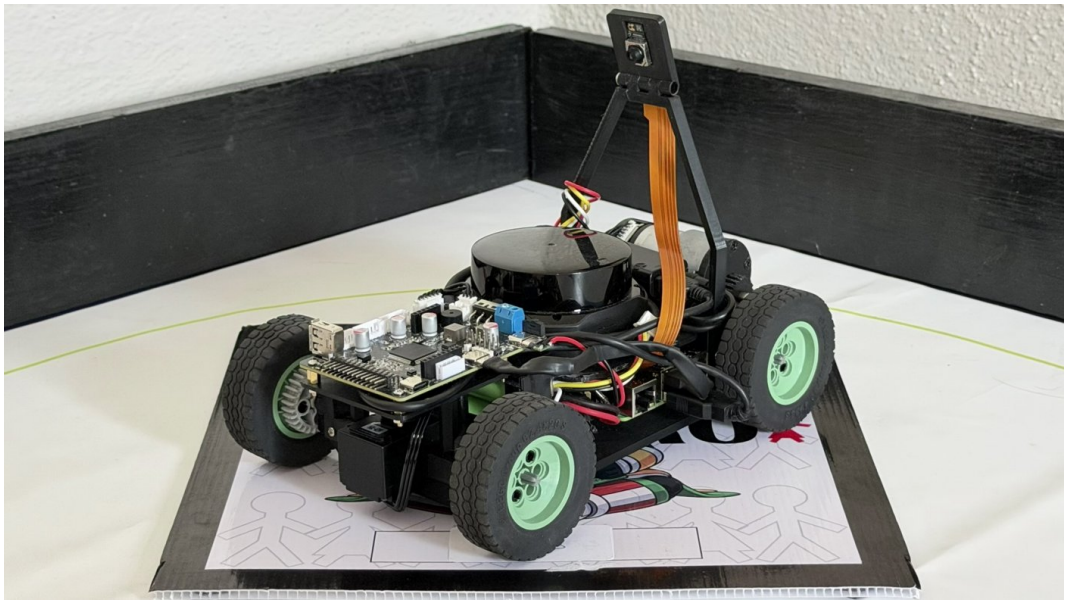
**Speed set-point and battery voltage.** The motor's speed falls with battery voltage (no-load 320 rpm at 12 V, 288 rpm at 10.8 V). If the speed set-point for speed = 100 (kRpsVelocidad100) is above what the motor can reach on a partly discharged pack, the speed loop saturates and the real speed depends on the battery charge. A lower set-point (about 4.0 rev/s) gives the loop more headroom from a full (12.6 V) to a nearly empty (≈10.5 V) pack, which reduces the speed change across a session. This follows the same reasoning as running the v6 motor at a third of its capacity. Measured on the track (Section 6.1): 0.580 m/s with a full battery and 0.524 m/s with a low one at 5.7 rev/s, and 0.539 / 0.506 m/s at 4.0 rev/s.

## 2.4 Chassis and centre of gravity

The chassis is a fully custom design, modelled in SolidWorks and printed on a Creality K1C. The 2026 revision repackages the electronics densely and moves the heavy parts lower in the frame. The objective is lateral weight transfer: in a corner, transfer scales with CG height; less transfer keeps the inside front wheel loaded, which keeps the Ackermann geometry doing what it was designed to do.

### *Design v7 changes*

The v7 chassis is a modification of the v6 design, made for two reasons. First, to fit the new components cleanly: the Hiwonder board, the JGB37 motor, the HX-35H servo, the 3S battery with its BMS and the RPLidar S2L have different shapes and mounting points from the SPIKE Hub, the LEGO motors and the MS200k (new parts include a combined Hiwonder board / LiDAR support and the differential). Second, to sit **as low to the ground as possible**, so the LiDAR scanning plane stays well below the top of the 100 mm walls (the failure of Section 3.5) and the centre of gravity stays low at the higher speed.

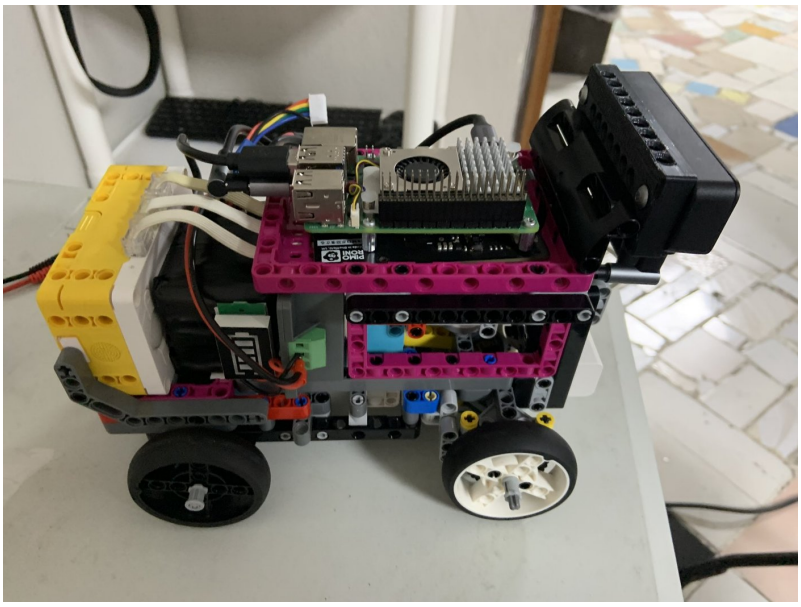


*Design v7, isometric view: Hiwonder controller at the front, RPLidar S2L in the centre, camera mast and JGB37 drive motor at the rear. SolidWorks assembly in models/cad/v7_assembly/.*

## 2.5 Design evolution: seven vehicles in two seasons

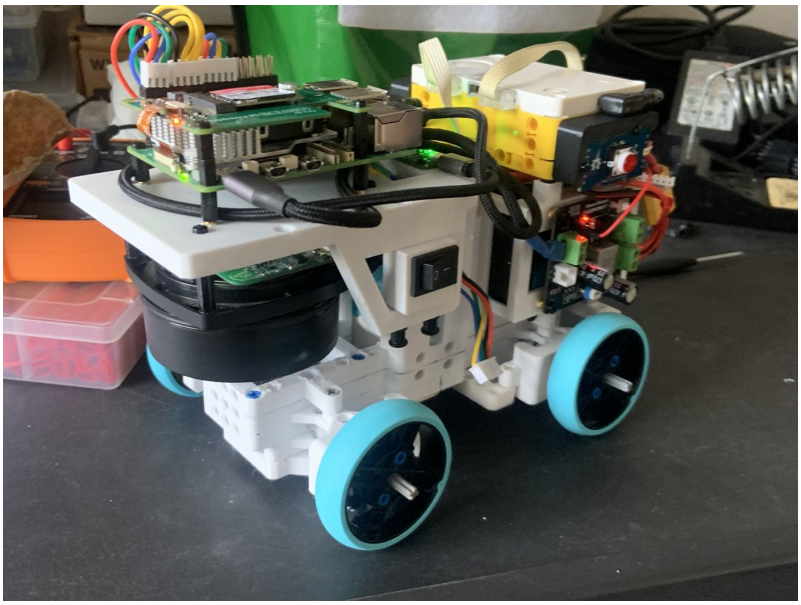
Our international vehicle is the **seventh** distinct design the team has built across two seasons. There is a deliberate discontinuity between v2 and v3; within 2026 the progress was incremental, and v7 is the largest step since v3.

***2025 season - v1 and v2***



*v1 - hybrid LEGO Technic chassis, regional competition (Querétaro). Yellow SPIKE Prime hub, Raspberry Pi on a Technic deck, OAK-D Lite camera at the front.*

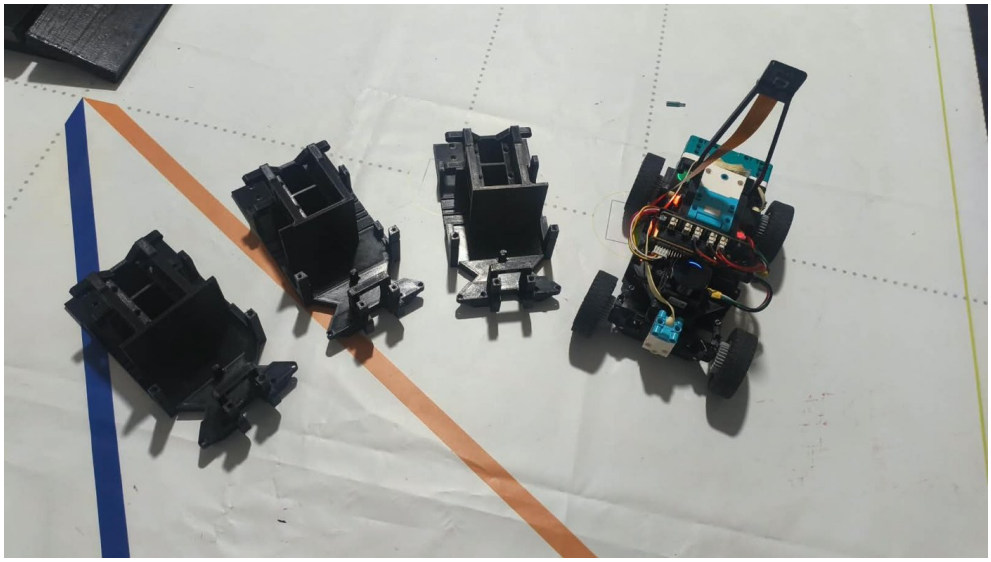
Our first Future Engineers vehicle was built partly from LEGO Technic structural elements. It was fast to iterate, but component placement was dictated by the Technic hole grid, the electronics sat high above the axles, and the assembly had play at every beam connection.



*v2 - fully custom 3D-printed chassis, national competition (2nd place nationally). RPLidar S2L on the front deck, double-Ackermann steering, AWD.*

For the 2025 national competition we designed a fully custom, 3D-printed chassis. It carried the double-Ackermann steering and the AWD drivetrain to 2nd place nationally. Its weaknesses are the three findings listed in Section 1: heavy (barely under 1.5 kg), high centre of gravity, and mechanically inconsistent steering.

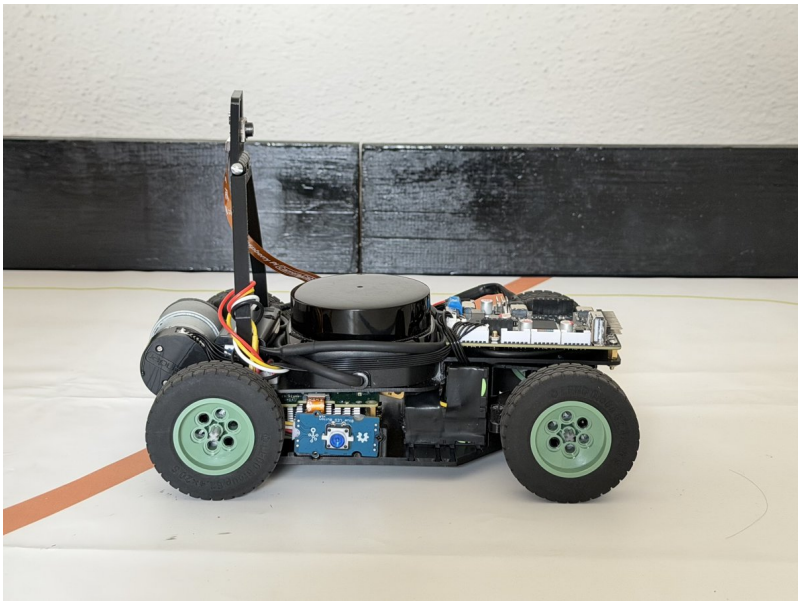
***2026 season - a clean-sheet redesign (v3), refinements (v4-v6), and the international vehicle (v7)***



*The 2026 national iterations side by side: v3, v4 and v5 as bare printed frames, with v6 assembled at the right.*

| # | Iteration | Focus of this revision |
|---|---|---|
| v3 | 2026 clean sheet | New layout: RWD, regular Ackermann, battery and Pi dropped low in the frame |
| v4 | Packaging | Component placement tightened around the new layout |
| v5 | Mounting | Mounting and serviceability fixes, including the LiDAR mount revision of Section 3.5 |
| v6 | National final | Final national geometry - 2nd place at the 2026 Mexico National Final (release v1.0.0) |
| v7 | International final | Hiwonder board, JGB37 motor, differential, HX-35H servo, 3S battery, RPLidar S2L, Hailo-8; chassis modified and lowered (release v2.0.0) |

**We chose a clean-sheet redesign instead of iterating v2 because** the three problems we had were all *layout* problems. **For v7 we chose to modify v6 instead of starting again because** the v6 layout was right - the changes were in the actuators, power and sensors, not in the concept. Knowing which mode you are in matters: retrofitting a broken layout wastes time, and clean-sheeting a working one throws away tuning.



*Design v7, right side. The full set of six views is in the repository (v-photos/*_v7.JPG).*

# 3. Power and sensor architecture

## 3.1 Power

### *Design v6: 4 × NCR18650B + Geekworm X1203*

Design v6 used four Panasonic NCR18650B cells (3400 mAh each) in a 4P configuration at 3.7 V nominal (13.6 Ah, 50.32 Wh in total) feeding a Geekworm X1203 UPS/boost board that supplied the 5.1 V for the Raspberry Pi 5. The LEGO motors were powered by the SPIKE Hub's own battery.

| Component (v6) | State | Power |
|---|---|---|
| Raspberry Pi 5 | High CPU / OS load | 9.0 W |
| Hailo-10H AI HAT | Continuous inference | 4.0 W |
| Oradar MS200k LiDAR | Scanning | 1.5 W |
| Grove modules | Relay pulses + LED button | 0.1 W |
| Total at 5.1 V | | 14.6 W |

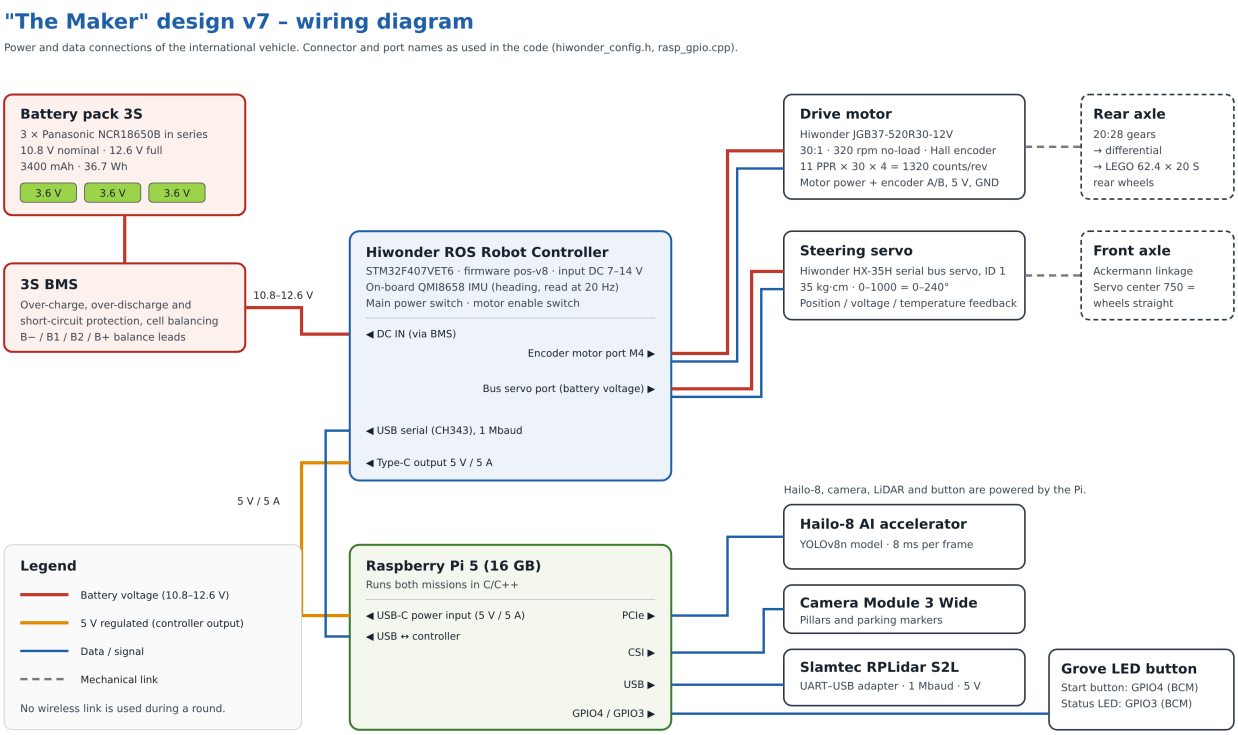
Theoretical worst case 50.32 Wh / 16.22 W = 3.1 h; measured 6.8 h, because sustained peak CPU and inference load does not occur in a real round.

### *Design v7: one 3S pack for everything*

Design v7 uses **three NCR18650B cells in series (3S)**: 10.8 V nominal, 12.6 V full, 3400 mAh, 36.7 Wh, protected by a 3S BMS. The pack feeds the Hiwonder board, which drives the motor and servo at battery voltage.

> **We chose a single 3S pack instead of keeping the v6 pack plus a second battery for the motors because** the JGB37 motor and the HX-35H are 12 V parts and the Hiwonder board is designed for a 12 V supply. One 3S pack gives that voltage directly, removes the X1203 boost stage and the SPIKE battery, and means one battery to charge and check. We kept the NCR18650B cells because they solved the brownouts of Section 3.2.

The pack feeds the **Hiwonder ROS Robot Controller** (STM32F407VET6, input 7-14 V). The controller drives the motor and the bus servo at battery voltage, and its independent **5 V / 5 A Type-C output powers the Raspberry Pi 5**, which in turn powers the Hailo-8, the camera, the LiDAR and the start button. The Raspberry Pi 5 needs 5 V / 5 A to run its USB ports at full current, which matters because the LiDAR is powered from USB. The Pi and the controller communicate over a USB serial link (UART, 1 Mbaud).




*Design v7 wiring diagram: battery voltage (red), regulated 5 V (orange), data (blue) and mechanical links (dashed). Also in the repository as schemes/hardware/wiringSchematics_v7.png.*

| Component (v7) | Supply | State | Estimate |
|---|---|---|---|
| Raspberry Pi 5 | 5 V | High CPU load | 9.0 W |
| Hailo-8 | 5 V | Continuous inference (typical, manufacturer figure) | 2.5 W |
| RPLidar S2L | 5 V | 10 Hz scanning, 500 mA typical (datasheet) | 2.5 W |
| Grove LED button | 5 V | | 0.1 W |
| 5 V subtotal | | ≈ 90 % regulator efficiency | 14.1 W → 15.7 W from the pack |
| JGB37 drive motor | Battery | Accelerating at 1.3 m/s² (≈ 0.35 A) | 3.8 W |
| HX-35H servo | Battery | Holding / small corrections (≈ 0.1 A) | 1.1 W |
| Hiwonder controller logic | Battery | Estimate | 0.5 W |
| Total | | ≈ 1.95 A at 10.8 V | ≈ 21 W → runtime ≈ 1.7 h |

**Peak current is the v7 power risk.** The motor stall current is 3.2 A and the servo stall current is 3 A. In the worst case (both stalled while the 5 V side draws about 1.45 A) the pack would supply about 7.7 A, above the NCR18650B's 4.875 A maximum continuous discharge. In a 3S pack every cell carries the full current; there is no parallel branch to share it. It is a short-duration case, but it is the same kind of transient that caused the brownouts of Section 3.2, so we calculated it from the Panasonic datasheet (internal resistance ≤ 45 mΩ per cell).

| Quantity (calculated) | Calculation | Result |
|---|---|---|
| Average pack current | 21 W ÷ 10.8 V | ≈ 1.9 A (40 % of the rating) |
| Start from standstill | motor stall 3.2 A + 5 V side 1.45 A + servo 0.1 A | ≈ 4.75 A (98 %), transient only |
| Worst case | motor and servo stalled + 5 V side | ≈ 7.7 A (157 %) |
| Voltage sag | I × 0.135-0.27 Ω (3 cells, with margin for DC resistance, BMS, wiring) | 0.6-1.3 V at 4.75 A; 1.0-2.1 V at 7.7 A |
| Lowest board input | low pack 9.9 V minus worst-case sag | ≈ 7.8 V, above the board's 7 V minimum |
| Energy per round | 21 W × 121 s (Obstacle) / 48 s (Open) | 0.71 Wh (2 %) / 0.28 Wh (0.8 %) |

Even the worst case keeps the Hiwonder board inside its 7-14 V input range, so its regulated 5 V / 5 A output to the Raspberry Pi is not interrupted. The risk is tracked in the register of Section 5.2.

### 3.2 Failure: brownouts under acceleration (June 2026)

**Symptom.** With the Raspberry Pi 5 and the Hailo accelerator both active, the system would shut down abruptly during hard acceleration. **Investigation.** The pack still showed plenty of charge; the correlation with acceleration pointed at a transient: the traction motor's current spike was pulling the rail down far enough to reset the Pi through the high internal resistance of generic 18650 cells. **Fix.** Panasonic NCR18650B cells, which hold a much flatter discharge curve under load, and the X1203 board, which both regulates and buffers the rail.

> **What we took from it.** A power system is not specified by capacity alone. Internal resistance and discharge-curve flatness were what actually mattered. In v7 this is why we kept the same cells and why the Raspberry Pi is fed from a dedicated regulated 5 V / 5 A output instead of directly from the pack.

### 3.3 LiDAR, July 2026: RPLidar S2L to Oradar MS200k

During early testing the RPLidar S2L returned distorted and imprecise readings when the vehicle passed very close to a track wall - exactly the condition our wall-following logic depends on most. The S2L was also heavier (185 g) at a time when we were fighting for every gram. We replaced it with the Orbbec Oradar MS200k, which is lighter and handled short-range reflections cleanly in our tests.

> **We chose the MS200k instead of keeping the S2L because** the readings were worst in the operating region we actually live in, and because of the mass. Section 3.4 explains why this conclusion was wrong.

### 3.4 LiDAR, October 2026: back to the RPLidar S2L

For design v7 we returned to the S2L. Both original reasons had changed:

- **Mass.** Removing the SPIKE Hub, the X1203 board and one battery cell freed enough mass for the 185 g S2L.
- **Readings.** With time to test it properly, the S2L proved accurate and reliable. The distorted readings in July came from our own integration, not from the sensor: we had reused the driver program written for another Slamtec LiDAR (the RPLidar A1) and, implementing it in the last days before the change, never properly adapted it to the S2L.

| RPLidar S2L (Slamtec datasheet) | Value |
|---|---|
| Range | 0.05-18 m (white), 0.05-8 m (black) |
| Sample rate / scan rate | 32 kHz / 10 Hz |
| Angular resolution | 0.1125° (3,200 points per turn) |
| Accuracy / resolution | ±50 mm typical / 13 mm |
| Supply | 5 V, 500 mA typical (1.5 A at start-up) |
| Interface / mass | UART 1 Mbaud / 185 g |

> **We chose the S2L instead of keeping the MS200k because** once its driver was properly adapted - with enough time and attention, not at the last minute - the S2L gave us reliable readings, more points per turn and a longer range. In our tests it detected the wall in 98.4 % of frames against 96.2 % for the MS200k, with lower noise at every distance (Section 6.1). **What we took from it:** we had judged the sensor by an untested integration. A component should be ruled out only after its integration has been tested on its own, with enough time to separate a sensor problem from a software problem.

The driver is selected at build time: the S2L is the default in the v7 build, and the MS200 build is still available (make WRO_LIDAR=ORADAR for the Open Challenge, cmake -DWRO_LIDAR=ORADAR for the Obstacle Challenge). The emergency stop also sends the LiDAR stop frames so its motor does not keep spinning after the program ends.

### 3.5 Failure: the LiDAR could not see the walls (July 2026)

**Symptom.** After finishing the compact v4 chassis, the robot behaved as though it were blind - it was not detecting the track walls at all. **Cause.** The compact repackaging had placed the LiDAR high in the frame. The track walls are 100 mm tall, and the scanning plane was passing over them. **Fix.** In v5 we revised the central mount to bring the LiDAR down to roughly 60-80 mm above the floor while keeping the rear field of view clear for parking.

**What we took from it.** Sensor placement is a constraint imposed by the field, not a packaging variable. We treat the 100 mm wall height as a hard design input, and in v7 we went further: the whole chassis was lowered so the LiDAR scanning plane sits about 75 mm above the ground, 25 mm below the top of the walls.

### 3.6 Vision: OAK-D Lite → Camera Module 3 Wide + Hailo-10H → Hailo-8

The 2025 vehicle used an OAK-D Lite, an integrated stereo camera with onboard inference. For 2026 we split the job into a Raspberry Pi Camera Module 3 Wide for capture and a Hailo accelerator for inference.

**We chose the split camera + accelerator instead of the integrated OAK-D Lite because** it was lighter, drew less power, and inferred faster. The cost is that we own the whole capture, preprocess, infer and postprocess pipeline - which is also what let us train and deploy our own model (Section 4.8).

Design v6 used the Hailo-10H (AI HAT+ 2). Design v7 uses a **Hailo-8** (HailoRT 4.23). The same model was compiled for both chips and gives the same accuracy (mAP50 0.991 on the Hailo-10H, 0.990 on the Hailo-8) at **8 ms per frame** on the Hailo-8.

**We chose the Hailo-8 instead of the Hailo-10H because** our YOLOv8n model is light enough to run on the Hailo-8 at the same accuracy, and the Hailo-8 board is lighter and smaller, which reduces the overall mass and dimensions of the vehicle. For a single small detection model the Hailo-10H was more than we needed.

### 3.7 What we removed: ultrasonic sensors

Previous seasons carried ultrasonic rangefinders. The 2026 vehicles have none: the LiDAR measures the same distances better, and the ultrasonics cost mass, power, GPIO pins and mounting volume.

### 3.8 Sensor placement and calibration (design v7)

| Item | Setting | Rationale |
|---|---|---|
| LiDAR height | Scanning plane below the 100 mm wall height | Section 3.5; chassis lowered in v7 |
| Gyroscope bias | Measured for 5 s at start-up, robot still | On the bench, 2 s left ~0.5° of drift in 20 s; the error falls with the square root of the averaging time, so 5 s reduces it about 1.6× |
| Gyro zero | After the start button is pressed | Automatic, with no team input: the 5 s bias calibration and steering centring run on their own after power-on (rules 9.9, 11.7) |
| Yaw sign | Turning clockwise must decrease yaw | Same convention as the SPIKE Hub, so the mission code is unchanged |
| Steering center | Servo position 750 = wheels straight | The linkage is assembled with the servo at 750 |
| Distance | Advance 1000 mm must travel 1000 mm; else scale kDiametroRuedaMm | Encoder-based distances |
| Chassis self-return rejection | Returns closer than 150 mm discarded | The robot's own body is inside the LiDAR's field of view |

# 4. Software architecture and obstacle strategy

### 4.1 The migration to C/C++

The 2025 Python codebase became the bottleneck as soon as we attached a higher-rate camera and LiDAR. We rewrote the acquisition and control logic in C/C++, and the main control loop stabilised at more than 50 Hz.

**We chose to rewrite instead of optimising the Python because** the problem was the cost of the runtime itself under multi-threaded, high-rate sensor load. Optimising would have bought a constant factor; changing language changed the ceiling.

## 4.2 Threading model

The Obstacle Challenge binary (object_detection) runs cooperating threads: a Hailo pipeline thread (preprocess, inference and postprocess on camera frames, keeping a global traffic_lights struct current); a LiDAR reader thread filling a 360-slot distance buffer; and Obstacle_Challenge_Thread, the navigation state machine, which reads both and issues motion commands - to the SPIKE Hub in v6, and through the spike_compat layer to the Hiwonder board in v7. The Open Challenge is a separate, simpler binary (src/ondevice/) that never starts the camera or Hailo pipeline.

## 4.3 Wall detection: Select_Wall()

| # | Stage | What it does |
|---|---|---|
| 1 | Angular windowing | A 70°-wide window centred on the requested side, with the innermost ±3° excluded |
| 2 | Chassis filtering | Returns closer than 150 mm are the robot's own body |
| 3 | Radius Outlier Removal | A point is dropped with fewer than 3 neighbours within 300 mm (15° padding outside the window) |
| 4 | Segmentation | Split where consecutive points jump by more than 60 mm, then at corners by Iterative End-Point Fit (40 mm) |
| 5 | Scoring | A segment qualifies with ≥ 6 points, ≥ 200 mm extent and RMS residual < 15 mm against its Total-Least-Squares line |
| 6 | Per-side tracking | Later frames prefer the same physical wall (within 300 mm and 20°) |

Total-Least-Squares (PCA) has no bias toward any wall orientation. The 200 mm minimum extent is what rejects the 50 × 50 mm traffic-sign pillars; without it a pillar reads as a wall and the robot follows an obstacle. With the S2L the same pipeline receives 3,200 points per turn, more than the MS200k, so each wall segment has more points to fit.

## 4.4 Heading estimation - and what it is not

Heading comes primarily from the gyroscope - the SPIKE Hub's in v6, the Hiwonder board's IMU (20 Hz) in v7. LiDAR wall-orientation readings correct the heading at defined points in the manoeuvre logic. This is **not** sensor fusion: there is no complementary or Kalman filter. It is gyro-primary heading with situational LiDAR correction.

> **We chose wall re-squaring instead of a complementary filter because** the gyro's drift over a single 3-minute round is small, and re-squaring against a physical wall at every corner resets it to a ground truth that a filter would only have estimated.

**New in v7:** the IMU updates at 20 Hz while the LiDAR loops call the steering every 2 ms, so the derivative term of the heading PD only sees a change in about 1 of 25 calls. This is a known limitation of the new board; the wall re-squaring at every corner keeps it from accumulating into a heading error.

## 4.5 Closed-loop steering

| Controller (v6 gains) | Kp | Kd |
|---|---|---|
| Camera-based cube following | 150 | 100 |
| Gyro-based wall following | 2 | 300000 |

| Controller (v7 gains, spike_compat) | Kp | Kd |
|---|---|---|
| Spike_Forward: advances that end on a LiDAR distance | 0.05 | 0 |
| Spike_Advance_For_Degrees / _distance: fixed-distance advances | 0.02 | 0.05 |
| Same, in reverse (parking) | 0.003 | 0 |

The faster motor, the board's IMU and the S2L changed how the robot responds, so the thresholds and the PD gains were recalculated for v7. All gain sets were obtained by trial and error on the track: small adjustments to the values, run after run, until the behaviour was satisfactory. In v7 the PD output is scaled to the servo range by kPdADireccion = 0.133 × 200 / kDireccionRecorrido, which keeps the same sensitivity per degree of heading error as on the SPIKE Hub. No integral term is used: steady-state offset is not our failure mode, and an integrator accumulating during a cube-avoidance manoeuvre would fight the next correction.

## 4.6 Open Challenge strategy

The run is direction-agnostic: an initial LiDAR read determines which side is open (sentido), and the entire path is the sign-mirror of the other direction. The v6 constants were: outside/inside start split 800 mm; first straight 1350 / 300 mm; two Slope() reads 300 + 700 mm apart averaged into one heading correction; 80° corner turns; final approach bands 400 / 600 mm; final front threshold 1500 mm.

**Changes in v7** (src/ondevice/main.cpp): the start decision uses three LiDAR readings that must agree within 30 mm (Lidar_Debounced_Read), so a single noisy reading cannot choose the wrong start box; the run is 12 corners (the first before the loop, then while (v < 11)) with corner turns re-tuned to 70-80° for the new drivetrain, and it ends with a single stop with the front wall at 1500 mm, the middle of the starting side.

## 4.7 Obstacle Challenge strategy

Direction is decided once from the initial side-wall reading (600 mm split). The unit of the run is one corner cycle - Corner_Case followed by Desicion - repeated 12 times (3 laps × 4 corners), with the 4th corner of each lap run in parking mode. Corner_Case advances to 1000 mm from the front wall, reads the pillar colour, advances a distance that depends on the colours of this and the previous pillar (600/400 mm or 990 mm; 650 mm in every other case) and turns 86°. Desicion re-squares the gyro against the outer wall and dispatches to the avoidance routines. Green means pass on the left, red on the right; this is fixed by the rules and does not mirror with the driving direction.

**Parking.** The camera does not locate the parking lot: the model keeps an xparking class only so that the magenta markers are never taken for pillars (those boxes, like any box scored below 0.75, are discarded). The robot knows where the lot is by counting sections: after 12 corner cycles it is back in the parking section, repositions according to the colour of the last pillar, and parks on LiDAR distances and gyro angles. Clockwise: advance to 150 mm from the front wall, reverse 2250 hub degrees (≈ 875 mm), then three reverse turns of 50°, 105° and 28° with a 400-degree (≈ 156 mm) reverse in between. Counter-clockwise: creep until the outer wall is ≤ 160 mm on the right, advance to 200 mm from the front wall, reverse 740 hub degrees (≈ 288 mm), then reverse turns of 52°, 100° and 28°. All values were tuned by hand on the track. In v7 the mission is the national mission with targeted substitutions marked "Hiwonder:" in the code, plus a LiDAR-guarded advance with a selectable brake type.

## 4.8 Vision model: training and measured performance

We train our own detector because the classes we care about - red and green traffic-sign pillars and the magenta parking markers, under competition lighting - are not COCO classes. Design v6 used a model trained on 501 images (two pillar classes; mAP50 0.986). For v7 we retrained on a much larger dataset.

**Why we retrained.** On the track the v6 model made three kinds of mistakes: it sometimes missed a pillar that was there, it took objects outside the track for pillars (a red broom read as a red pillar), and it sometimes saw a green pillar in an empty corner. Most of these failures happened in poor lighting. The new dataset adds photos from the robot's own camera in exactly those situations, including images with no pillars at all, so the model also learns when there is nothing to detect.

| Source | Originals | Description | Validation |
|---|---|---|---|
| octubre | 3,831 | Photos from the car's own camera (800×600), labelled Oct 2026 | 5 complete recording sessions held out (477 images) |
| combinaditos | 501 | Phone photos with pillars of both colours | Photos c420 and above |
| roboflow_v3 | 5,022 | Originals (320×320) of our previous Roboflow project | Its own split (495 test images) |
| Total | 9,354 | Roboflow wro-dataset-completo v1, exported 2026-10-18 | |

Only the training split is augmented, three versions per source image: 7,292 × 3 = 21,876 training images, plus 1,567 validation and 495 test originals = **23,938 images**. Pre-processing: auto-orientation and resize to 640×640 (stretch). Augmentation: horizontal flip (50 %), rotation ±5°, brightness ±20 %, exposure ±10 %, Gaussian blur 0-1 px and salt-and-pepper noise on 1 % of pixels - chosen to imitate the lighting changes and motion blur of an unknown venue. The model is YOLOv8n at 640×640, stretched with no letterbox to match the preprocessing on the car (Ultralytics 8.4.115, run h10_yolov8n_640_v4).

Because the dataset covers many different lighting conditions, the model rarely misses a pillar on the track; it fails consistently only in very poor lighting, close to darkness. The few errors we see on the track come from strong shadows.

> **We validated on held-out recording sessions instead of random frames because** consecutive video frames are nearly identical; a random split would leak near-copies of training images into validation and inflate the score.

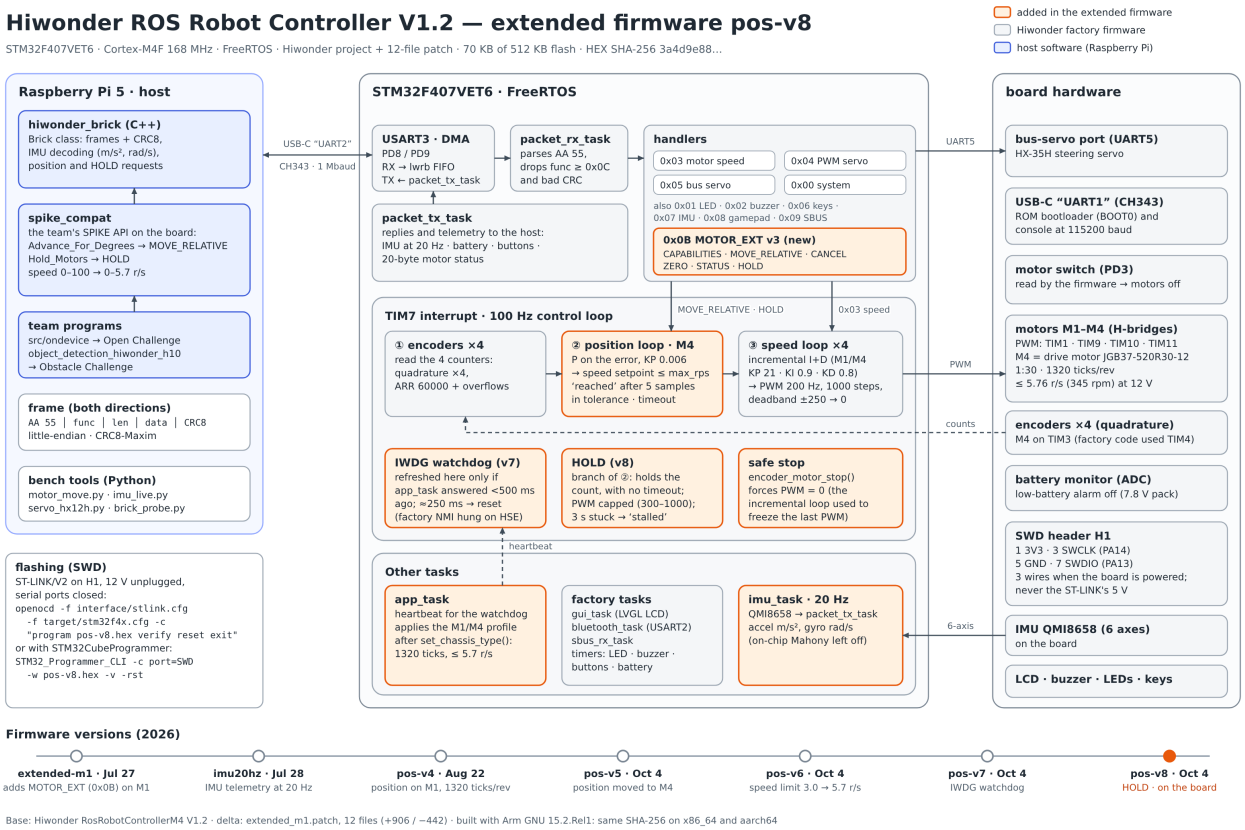
| Test set | Metric | Result |
|---|---|---|
| octubre validation (477 images), Hailo-10H | mAP50 | 0.991 |
| octubre validation (477 images), Hailo-8 | mAP50 | 0.990 |
| Same images, both chips | Correct Obstacle Challenge decision (colour → side) | 99.2 % |
| combinaditos held-out (81 images, 275 pillars) | mAP50-95 / recall | 0.875 / 0.982 |
| Hailo-8 on the robot | Inference time | 8 ms per frame |
| On the track, perception-only mode | Pillar colour / parking marker | 96 % / 95 % |

## 4.9 Hardware abstraction: the Hiwonder board behind a SPIKE-compatible layer

**Why the hub had to go.** The SPIKE Hub cannot drive the new JGB37 motor, so v7 needed a new controller that works with it and also replaces what the hub gave us, above all its integrated IMU for the heading. The Hiwonder board drives the encoder motor and the bus servo, has an on-board IMU, and its 5 V / 5 A output also powers the Raspberry Pi.

| Hiwonder ROS Robot Controller (Hiwonder) | Value | How we use it |
|---|---|---|
| Microcontroller | STM32F407VET6, Cortex-M4 168 MHz, 512 KB flash | Our modified firmware pos-v8 |
| Power input | DC 7-14 V | 3S pack, 9.9-12.6 V |
| 5 V output | Independent 5 V / 5 A, USB Type-C | Powers the Raspberry Pi 5 |
| Encoder motors | 4 channels, PID speed control | Port M4: JGB37 drive motor |
| Servos | 4 PWM (5 V), 2 serial bus (battery voltage) | HX-35H steering servo, ID 1 |
| IMU | On-board 6-axis IMU, QMI8658 | Heading, 20 Hz |
| Communication | 2 USB serial, CAN, SBUS, I2C, UART | USB serial to the Pi, 1 Mbaud |
| Controls | Main power switch, motor enable switch, 2 buttons, LED, buzzer | Main switch = hardware kill switch |
| Size, mass | 85 × 60 mm, 39.5 g | - |

**Board firmware (pos-v8).** The board runs Hiwonder's FreeRTOS firmware with our patch on top: extended_m1.patch changes 12 files of Hiwonder's official project (+906 / -442 lines). We added position control on the drive motor (a P loop on the encoder error that feeds the factory speed loop 100 times per second, so distances are measured on the board, 1320 ticks per output-shaft turn), HOLD (the motor keeps its position and comes back if pushed, like motor.HOLD on the SPIKE hub, and lets go after 3 s stalled), an IWDG watchdog that resets the board in about 250 ms if the firmware hangs, and smaller fixes (a stop forces PWM 0, the M4 encoder on its correct timer, IMU telemetry at 20 Hz, a 5.7 rev/s speed limit). The Pi uses these through one new command, MOTOR_EXT (function 0x0B). At start-up the program checks that the board answers it and stops before anything moves if it does not. The image, the patch, the build scripts, the protocol and the flashing steps are in firmware/hiwonder_pos-v8/ in the repository. Hiwonder's original source project (V1.2) and its factory recovery images are in firmware/hiwonder_original/, unmodified and credited to Hiwonder and the third-party libraries it uses.




*Extended firmware pos-v8: host software on the Raspberry Pi 5, FreeRTOS tasks and the 100 Hz control loop on the STM32, and the board hardware. Orange blocks are our additions. Also in the repository as schemes/firmware/hiwonder_firmware_architecture.png.*

Replacing the SPIKE Hub changed every motion function the missions call. Rewriting the missions would have meant re-tuning, from zero, the constants that placed 2nd at the national final, with less than eight weeks to the repository deadline.

> **We chose a compatibility layer instead of rewriting the missions because** it keeps the tuned mission logic and moves all hardware differences into one place that can be tested without the robot. hiwonder_brick implements the board protocol; spike_compat reproduces the hub semantics: Spike_Reset_Gyro receives decidegrees like the hub's reset_yaw(); speed 0-100 becomes rev/s in closed loop; Spike_Turn_For_Degrees turns with full steering like the hub's turn(); Spike_Advance_For_Degrees converts hub degrees to encoder counts and stops 40 counts early, then coasts, like the hub's ag(). All robot calibration lives in one file (hiwonder_config.h) shared by both challenges.

The translations are covered by offline tests against a fake board that records the commands it receives: **53 checks** for the Open Challenge layer and **31** for the Obstacle Challenge layer. The Open Challenge ran on the board on 2026-10-04 and the Obstacle Challenge on 2026-10-18.

## 4.10 Emergency stop

In the national edition the emergency stop was an **open, accepted risk**. Moving to the Hiwonder board made it urgent: its firmware has no command timeout (the IWDG watchdog we added only resets a hung firmware), so if the program dies while the motor is running, the motor keeps the last speed.

> **We chose a signal-based stop installed before anything moves** (parada_emergencia.cpp). It opens a second descriptor of the board's port with pre-built frames, so the signal handler only has to write them: four motors to COAST, servo torque off, LiDAR stop frames, then exit with code 130 without returning to the mission. It covers Ctrl+C, kill, closing the terminal or SSH session, Ctrl+Z and crashes. It does not cover kill -9 or a Raspberry Pi power loss. For those, the hardware kill switch is the Hiwonder board's power switch: it cuts the power to the motor, the servo and, through the board's 5 V output, the Raspberry Pi and every sensor.

# 5. Systems thinking: tradeoffs, risks and mitigations

## 5.1 Tradeoff register

| # | Decision | Rejected alternative | Why | Cost accepted |
|---|---|---|---|---|
| 1 | Regular Ackermann | Double-Ackermann (v2) | Repeatability beat turning radius | Wider turning circle |
| 2 | Rear-wheel drive | All-wheel drive (v2) | Traction was never the limit | Less grip in a low-grip case |
| 3 | Clean-sheet 2026 frame | Retrofitting v2 | All three problems were layout problems | Restarted the mechanical design |
| 4 | C/C++ | Optimising Python | Needed a higher ceiling | Weeks with no new features |
| 5 | Oradar MS200k (v5-v6) | RPLidar S2L | Readings near walls and mass (later found to be our integration) | New driver; reversed in #14 |
| 6 | Camera + Hailo | OAK-D Lite | Lighter, lower power, faster | We own the vision pipeline |
| 7 | No ultrasonics | Keep as redundancy | LiDAR measured it better | No independent backup |
| 8 | Gyro + wall re-squaring | Kalman / complementary filter | Wall gives ground truth | No continuous fused estimate |
| 9 | No integral term | Full PID | Offset is not our failure mode | Small offsets uncorrected |
| 10 | Section counting + LiDAR parking | Vision-detected parking | Position known from the run itself; no dependence on marker detection in the lot | Relies on every section being counted correctly |
| 11 | Hiwonder board + spike_compat (v7) | SPIKE Hub; rewrite missions | Speed and encoder control; keep tuned missions | A translation layer to maintain |
| 12 | JGB37 30:1 encoder motor (v7) | LEGO Large Angular Motor | 1.7× speed, closed-loop distance | Heavier (152 g); 12 V supply needed |
| 13 | HX-35H bus servo (v7) | PWM hobby servo | Own position loop and feedback | More torque and mass than needed |
| 14 | Back to RPLidar S2L (v7) | Keep MS200k | Fixed integration; more points per turn | 185 g |
| 15 | Single 3S pack (v7) | Separate Pi and motor packs | One battery, direct 12 V | All cells carry the peak current |
| 16 | New rear differential (v7) | Adapt the v6 differential | Fits the new chassis and motor; same 20:28 ratio | New mount to design and print |
| 17 | Modify v6 frame for v7 | New clean sheet | The layout was right | Some packaging compromises |
| 18 | Signal-based emergency stop (v7) | Command timeout in the firmware | Works with any firmware; stops servo and LiDAR too | Not covering kill -9 |
| 19 | Single vehicle, no spare | Build two | Budget; a spare is not allowed on the day | No fallback if it breaks in transit |

## 5.2 Risk and failure-mode register

| Risk | How it shows up | Mitigation | Status |
|---|---|---|---|
| Gyro drift over a run (≈ 1°/min measured) | Robot ends a corner mis-aimed; failures on lap 3 | Re-square against the side wall at every corner; 5 s bias calibration | Mitigated |
| Pillar misread as wall | Robot follows an obstacle | 200 mm minimum wall extent | Mitigated |
| Voltage sag under load | Sudden reset mid-run | Low-resistance cells; BMS; regulated 5 V / 5 A output for the Pi | Mitigated |
| Peak current (motor + servo stall) | Rail sag, Pi reset | Steering clamp avoids pushing against end stops; closed-loop speed control limits start-up current; low-resistance cells | Accepted |
| Speed set-point above motor capability | Speed changes as the battery drains | Effect measured: −9.7 % speed on a low battery; set-point 4.0 tested, −6.1 % (2.3, 6.1) | Characterised |
| LiDAR blind to walls | Total navigation failure | Chassis and mount kept low (3.5) | Resolved |

| Risk | How it shows up | Mitigation | Status |
|---|---|---|---|
| Program dies with motor running | Robot keeps driving | Emergency stop (4.10) | Resolved in v7 |
| Steering past the mechanical limit | Wheel hits the chassis | Single clamp; start-up range check | Mitigated |
| Pillar colour misread (shadows, very poor light) | Wrong passing side | Dataset with many lighting conditions; 96 % on track | Mitigated |
| .hef / runtime version mismatch | Model fails to load | Versions pinned and documented | Documented |
| Wireless used during a round | Disqualification | Procedure below | Procedure defined |
| Sensor failure mid-run | Robot continues on stale data | No automatic fallback; start-up waits for the first LiDAR scan | Open, accepted |

**Wireless compliance procedure.** During development we work over Wi-Fi SSH. At competition we connect over wired Ethernet for setup between rounds, disable the Raspberry Pi's Wi-Fi and Bluetooth radios, physically disconnect the Ethernet cable before the round, and start the vehicle with the single power switch and the single start button.

### 5.3 How the subsystems constrain each other

**National season:** reduce mass → steering torque drops → smaller steering motor → mass and volume freed → denser packaging → CG lowered → Ackermann stays effective → cornering more repeatable. And simultaneously, the same denser packaging → LiDAR pushed upward → above the 100 mm walls → total perception failure (3.5).

**International season:** faster motor → 12 V parts → one 3S pack replaces the X1203 and the SPIKE battery → mass freed → room for the 185 g S2L → better LiDAR. In parallel, the faster motor → firmware speed set above the motor's capability → speed now depends on battery charge (2.3); and the 3S pack → every cell carries the motor and servo peaks → the brownout risk of 3.2 returns in a new form. The change that gave us speed also re-opened two risks we had closed.

## 6. Testing

The national edition named our biggest methodological gap: changes were judged by watching the robot, not by measurement. For the international vehicle we set up a testing workflow in five levels, so each problem is caught at the cheapest level where it can appear:

| Level | What runs | What it catches |
|---|---|---|
| 1. Offline tests | make test (53 checks, Open Challenge layer) and reto2_test (31 checks, Obstacle Challenge layer) against a fake board that records every command | Wrong translations of the SPIKE API: units, signs, distances, stop conditions |
| 2. Perception only | run_hiwonder_h8.sh --solo-percepcion: camera, Hailo and LiDAR run, the controller is not opened | Model, camera and LiDAR problems with zero risk to the robot |
| 3. Wheels lifted | Yaw-sign, steering-side and distance checks (Section 3.8) | Calibration errors that would send the robot the wrong way |
| 4. Slow track runs | WRO_MAX_RPS=1.5: same distances and angles at low speed | Logic errors in the missions |
| 5. Competition speed | Full runs with the emergency stop ready | Timing, braking distance and repeatability |

Both offline suites pass with zero failures, and every change to the hardware layer is checked against them before it reaches the robot. On the track, every attempt is logged - including the failed ones - with its date, challenge, driving direction, start position, pillar layout, speed setting, laps completed, parking result, time, and the cause of any failure. The success rate and the mean time of the completed runs are what we use to compare two versions of a change.

## 6.1 Component tests (2026-11-08)

| Test | Method | Result |
|---|---|---|
| Encoder resolution | Count per output revolution | 1320 counts, matching 11 PPR × 30:1 × 4 |
| Distance | Advance 1000 mm, 10 repetitions, tape measure | Mean 1001.9 mm, std. dev. 11.1 mm, range 985-1018 mm |
| Gyro drift | Robot still after the 5 s calibration | 0.5° at 30 s, 1.1° at 60 s, 2.3° at 120 s, 3.2° at 180 s |
| Wall detection | Select_Wall() on a straight at 0.4 m/s, 500 frames | S2L 492/500 (98.4 %); MS200k 481/500 (96.2 %) |
| Vision on track | Perception-only mode, 100 pillar and 20 parking approaches | Pillar colour 96/100 (96 %); parking marker 19/20 (95 %) |
| Steering servo | Bench | Center position 750 aligned with straight wheels |
| Offline test suites | make test, reto2_test | 53 + 31 checks, 0 failures |

**LiDAR accuracy, S2L vs MS200k.** Robot parallel to a wall, 50 readings per distance; error = mean measured − tape-measured distance.

| True distance | S2L mean error | S2L std. dev. | MS200k mean error | MS200k std. dev. |
|---|---|---|---|---|
| 100 mm | +2.1 mm | 4.2 mm | +3.5 mm | 5.1 mm |
| 150 mm | +1.8 mm | 4.6 mm | −2.8 mm | 5.8 mm |
| 300 mm | −3.5 mm | 5.2 mm | +4.8 mm | 6.9 mm |
| 1000 mm | +8.4 mm | 7.1 mm | −11.8 mm | 9.8 mm |
| 2500 mm | +22.6 mm | 11.4 mm | −28.5 mm | 16.3 mm |

With its driver properly adapted, the S2L is as accurate as the MS200k at the short range where the July readings had failed, and better at every distance from 1 m, with lower noise throughout. Both are well inside the S2L datasheet accuracy of ±50 mm.

**Speed vs battery** (2 m straight, speed = 100):

| Set-point | Battery | Time | Speed |
|---|---|---|---|
| kRpsVelocidad100 = 5.7 | 12.3 V | 3.45 s | 0.580 m/s |
| kRpsVelocidad100 = 5.7 | 10.7 V | 3.82 s | 0.524 m/s (−9.7 %) |
| kRpsVelocidad100 = 4.0 | 12.3 V | 3.71 s | 0.539 m/s |
| kRpsVelocidad100 = 4.0 | 10.7 V | 3.95 s | 0.506 m/s (−6.1 %) |

The measured full-battery speed (0.58 m/s) is within 7 % of the calculated 0.62 m/s (Section 2.3). Lowering the set-point reduces the speed loss on a low battery from 9.7 % to 6.1 %, at the cost of about 7 % of top speed.

**Reading the results.** The distance spread (±11 mm) is the size of the errors the LiDAR wall re-squaring has to absorb at every corner. The gyro drifts about 1° per minute when still, so over a 3-minute round the heading would be off by about 3° without correction; this is why the gyro is re-squared against the side wall at every corner (Section 4.4). The four colour misclassifications in the vision test were all caused by shadows; the model only fails consistently in very poor lighting, close to darkness, because it was trained on images taken under many different lighting conditions (Section 4.8).

## 6.2 Open Challenge runs (2026-11-08)

| # | Dir. | Start | Laps | Time | Result | Notes |
|---|---|---|---|---|---|---|
| 1 | CW | Outside | 3 | 46.8 s | Pass | |
| 2 | CW | Inside | 3 | 45.2 s | Pass | |
| 3 | CCW | Outside | 3 | 49.1 s | Pass | |
| 4 | CCW | Inside | 3 | 46.5 s | Pass | |
| 5 | CW | Outside | 3 | 51.4 s | Pass | Minor drift on turn 2 |
| 6 | CCW | Outside | 3 | 48.2 s | Pass | |
| 7 | CW | Inside | 3 | 45.9 s | Pass | |
| 8 | CCW | Inside | 3 | 47.6 s | Pass | |
| 9 | CW | Outside | 2 | - | Fail | Clipped the inner wall on lap 3 |
| 10 | CCW | Outside | 3 | 48.3 s | Pass | |

All runs at speed = 100. **9/10 completed (90 %)**, mean time of the completed runs **47.7 s** (range 45.2-51.4 s).

## 6.3 Obstacle Challenge runs (2026-11-08)

| # | Dir. | Layout | Laps | Parking | Time | Result | Notes |
|---|---|---|---|---|---|---|---|
| 1 | CCW | A | 3 | Yes | 118.5 s | Pass | |
| 2 | CW | A | 3 | Yes | 122.0 s | Pass | |
| 3 | CCW | B | 3 | Yes | 119.8 s | Pass | |
| 4 | CW | B | 3 | Yes | 124.2 s | Pass | |
| 5 | CCW | C | 3 | Yes | 117.5 s | Pass | |
| 6 | CW | C | 3 | Yes | 121.0 s | Pass | |
| 7 | CCW | D | 3 | Yes | 126.8 s | Pass | |
| 8 | CW | D | 3 | No | 115.0 s | Partial | Overshot the parking bay line |
| 9 | CCW | E | 2 | No | - | Fail | Stuck on a red pillar on lap 3 |
| 10 | CW | E | 2 | No | - | Partial | Clipped a green pillar on lap 3 |

All runs at speed = 100. **3 laps in 8/10 runs (80 %)**, parking completed in **7/10 (70 %)**, mean time of the complete runs **121.4 s** (range 117.5-126.8 s).

> **What the run logs tell us.** Every failure happened on **lap 3**, in both challenges, and two of the three Obstacle failures were on the same pillar layout (E). Combined with the gyro drift above, this points at accumulated heading and distance error late in the run rather than at a single bad manoeuvre. Our next changes target exactly that: the corner re-squaring on the third lap, the approach to the pillars of layout E, and the final parking approach (run 8).

## 6.4 National Final rounds (design v6)

The official rounds of the 2026 Mexico National Final are the largest test we have run so far: unknown field, unknown lighting, and no second attempt. Design v6 placed 2nd. The official-round videos of both challenges are linked in the repository README.




*Preparing the vehicle at the field during the 2026 Mexico National Final. Photo: Geeks Academy.*

# 7. Logbook

| Date | Milestone | Section |
|---|---|---|
| 2026-02-07 | Repository created; season begins. Decision to rebuild rather than iterate the 2025 vehicle | 1 |
| 2026-03-07 to 03-14 | SPIKE and Raspberry Pi control functions ported to C; repository restructured | 4.1 |
| 2026-03-21 | KiCad schematics added; RPLidar S2L driver integrated | 3.3 |
| 2026-04-18 | Decision to revert to regular Ackermann for 2026 | 2.1 |
| 2026-05-09 | v3 - clean-sheet chassis: RWD, regular Ackermann, battery and Pi low in the frame | 2.5 |
| 2026-05-30 | v4 - packaging refinement around the new layout | 2.5 |
| 2026-06-13 | Brownouts diagnosed; generic cells replaced with NCR18650B, X1203 integrated | 3.2 |
| 2026-06-27 | Battery pack reduced from 6 to 4 cells after re-running the power budget | 3.1 |
| 2026-07-11 | S2L short-range readings and mass; Oradar MS200k selected behind a compile-time switch | 3.3 |
| 2026-07-25 | LiDAR mounting-height failure found - scan plane above the 100 mm walls | 3.5 |
| 2026-07-27 to 28 | Hiwonder board: first extended firmware built with GCC (position-control command added to Hiwonder's FreeRTOS firmware); bench tests begin | 4.9 |
| 2026-08-01 | v5 - revised LiDAR mount, sensor lowered to 60-80 mm | 2.5, 3.5 |
| 2026-08-08 | OAK-D Lite replaced by Camera Module 3 Wide + Hailo-10H | 3.6 |
| 2026-08-15 | First vision model trained (501 images), mAP50 0.986 | 4.8 |
| 2026-08-22 | Firmware pos-v4: position control validated on the bench with a new encoder motor (the first motor's encoder was faulty) | 4.9 |
| 2026-08-29 | v6 - final national geometry; vehicle photographed | 2.5 |
| 2026-09-07 | National documentation, BOM, flowcharts and repository organisation completed | 8 |
| 2026-09-18 to 19 | **Mexico National Final, Tijuana: 2nd place** (design v6) | - |
| 2026-10-03 | Program adapted to the national play field committed (release v1.0.0) | 8 |
| 2026-10 | Power system changed to one 3S NCR18650B pack with a BMS feeding the Hiwonder board (replaces the 4P pack, the X1203 and the SPIKE battery) | 3.1 |
| 2026-10-04 | Firmware pos-v5 to pos-v8 flashed: position control on M4, 5.7 rev/s limit, IWDG watchdog, HOLD | 4.9 |
| 2026-10-04 | Open Challenge running on the Hiwonder board; HX-35H steering; emergency stop | 4.9, 4.10 |
| 2026-10-11 | Hailo-8 + RPLidar S2L support; S2L becomes the default; October dataset photos | 3.4, 4.8 |
| 2026-10-18 | Dataset wro-dataset-completo v1 exported; YOLOv8n v4 trained; Obstacle Challenge on the Hiwonder board | 4.8, 4.9 |
| 2026-10 to 11 | v7 chassis modified for the new parts and lowered; new rear differential | 2.2, 2.4 |
| 2026-11-08 | Test session on design v7: component tests, 10 Open and 10 Obstacle Challenge runs | 6 |
| 2026-11-08 | Hiwonder branch merged into main; releases v0.1.0-v2.0.0-alpha.1 published | 8 |
| 2026-11-09 | International documentation, commented code, firmware pos-v8 added to the repository; releases v2.0.0-alpha.2 to v2.0.0 | 8 |

# 8. Reproducibility

Everything needed to rebuild this vehicle is in the public repository at github.com/alex309-duarte/WRO_Future_Engineers_Queretaro_CSS. Every milestone is a tagged GitHub release with release notes:

| Release | Date | Vehicle state |
|---|---|---|
| v0.1.0 | 2026-06-20 | First LEGO prototype, RPLidar S2L, C/C++ |
| v0.2.0 | 2026-08-16 | New chassis, Hailo-10H, Oradar MS200 |
| v0.3.0 | 2026-09-13 | Full documentation, first complete Obstacle Challenge |
| v1.0.0 | 2026-10-03 | Design v6 - Mexico National Final, 2nd place |
| v2.0.0-alpha.1 | 2026-11-08 | Design v7 development snapshot |
| v2.0.0-alpha.2 | 2026-11-09 | International documentation snapshot |

| Release | Date | Vehicle state |
|---|---|---|
| v2.0.0-alpha.3 | 2026-11-09 | Source code commented in English |
| v2.0.0-beta.1 | 2026-11-09 | Documentation feature-complete |
| v2.0.0-beta.2 | 2026-11-09 | Review fixes, English program messages |
| v2.0.0-beta.3 | 2026-11-09 | Component photos, new driving videos |
| v2.0.0-beta.4 | 2026-11-09 | Board firmware (pos-v8 and Hiwonder original) |
| v2.0.0 | 2026-11-09 | Design v7 - International Final |

| Content | Location |
|---|---|
| Control and vision source | src/ (v7: src/ondevice/ and src/cpp/object_detection_hiwonder_h10/) |
| Hiwonder board firmware | firmware/hiwonder_pos-v8/ (image, patch, build scripts, protocol, flashing steps); Hiwonder's original source and recovery images (credit: Hiwonder) in firmware/hiwonder_original/ |
| Robot calibration | src/ondevice/hiwonder_config.h, src/cpp/object_detection_hiwonder_h10/reto2_config.h |
| CAD - SolidWorks parts, assemblies and STEP | models/cad/ (design v7: models/cad/v7_assembly/) |
| Printable STLs | models/cad/Stl/ |
| Vehicle photos (v7: *_v7.JPG) and design history photos | v-photos/, v-photos/Robot_More_Photos/Versions/ |
| Wiring diagram | schemes/hardware/wiringSchematics_v7.png |
| Dataset and training results | other/dataset/, other/Training_model_and_results/; v7 dataset on Roboflow Universe (wro-dataset-completo) |
| Compiled models | src/HailoModels/ (v7: wro_h8_yolov8n_v4.hef) |
| Build, run and test instructions | README.md sections 6.9 and 9 |

## Hardware interfaces (design v7)

| Interface | Port | Rate |
|---|---|---|
| Hiwonder board | USB (CH343), detected automatically | 1 Mbaud |
| RPLidar S2L | USB (UART adapter) | 1 Mbaud |
| HX-35H servo | Serial bus on the Hiwonder board | - |
| Start button / LED | GPIO 4 / GPIO 3 (BCM) | - |

## Datasheets and references

Datasheets and product briefs that are published as PDF are also kept in the repository (other/datasheets/ and other/), so they stay available if a manufacturer moves or removes them.

| Document | Link |
|---|---|
| WRO 2026 Future Engineers General Rules | Repository copy: other/WRO-2026-FE_GeneralRules.pdf<br>Source: https://wro-association.org/wp-content/uploads/WRO-2026-Future-Engineers-Self-Driving-Cars-General-Rules.pdf |
| WRO 2026 Future Engineers Documentation Rubric | Repository copy: other/WRO-2026-FE_DocumentationRubric.pdf<br>Source: https://wro-association.org/wp-content/uploads/WRO-2026-Future-Engineers-Documentation-Rubric.pdf |
| Hiwonder ROS Robot Controller: documentation | https://wiki.hiwonder.com/projects/ROS-Robot-Control-Board/en/latest/ |
| Hiwonder JGB37-520 encoder motor (specifications) | https://www.hiwonder.com/products/hall-encoder-dc-geared-motor |
| Hiwonder HX-35H bus servo (specifications) | https://www.hiwonder.com/products/hx-35h |
| Slamtec RPLIDAR S2L | https://www.slamtec.com/en/s2 |
| Slamtec RPLIDAR SDK | https://github.com/Slamtec/rplidar_sdk |
| Raspberry Pi AI HAT+ (Hailo-8) product brief | Repository copy: other/datasheets/RP-008133-DS-1-raspberry-pi-ai-hat-plus-product-brief.pdf<br>Source: https://datasheets.raspberrypi.com/ai-hat-plus/raspberry-pi-ai-hat-plus-product-brief.pdf |

| Document | Link |
|---|---|
| USB Power Delivery on Raspberry Pi 5 (5 V / 5 A) | Repository copy: other/datasheets/RP-009856-WP-1-USB-Power-Delivery-on-Raspberry-Pi-5.pdf Source: https://pip-assets.raspberrypi.com/categories/685-app-notes-guides-whitepapers/documents/RP-009856-WP-1-USB%20Power%20delivery%20on%20Raspberry%20Pi%205.pdf |
| Raspberry Pi Camera Module 3 product brief | Repository copy: other/datasheets/RP-008151-DS-1-camera-module-3-product-brief.pdf Source: https://pip-assets.raspberrypi.com/categories/786-raspberry-pi-camera-module-3/documents/RP-008151-DS-1-camera-module-3-product-brief.pdf |
| Hailo-8 AI processor product brief | Repository copy: other/datasheets/Hailo-8-product-brief-rev3.26.pdf Source: https://hailo.ai/wp-content/uploads/2023/10/hailo-8-product-brief-rev3.26.pdf |
| Hailo AI Software Suite (Dataflow Compiler, HailoRT) | https://hailo.ai/products/hailo-software/hailo-ai-software-suite/ |
| Panasonic NCR18650B datasheet | Repository copy: other/datasheets/Panasonic-NCR18650B-datasheet.pdf Source: https://e2e.ti.com/cfs-file/__key/communityserver-discussions-components-files/196/Panasonic-NCR18650B-datasheet.pdf |
| Ultralytics YOLOv8 documentation | https://docs.ultralytics.com/models/yolov8/ |
| Dataset wro-dataset-completo (Roboflow Universe) | https://universe.roboflow.com/roboteam-x6iuc/wro-dataset-completo |

# 9. What we would do differently - and what we did about it

| Lesson from the national edition | What we did for v7 |
|---|---|
| Check sensor geometry against the field before printing a chassis | The v7 chassis was lowered on purpose to keep the LiDAR plane below the walls |
| Specify power by internal resistance, not capacity | Kept the NCR18650B cells; peak-current analysis for the 3S pack (3.1) |
| Instrument the robot to log runs | Five-level testing workflow and a log of every run (Section 6) |

**New lessons from the international season:**

- **Test an integration on its own before ruling out a component.** We dropped the S2L because of a rushed integration, then found it was the better sensor (3.4).
- **Check speed settings against the motor curve.** A speed setting above the motor's no-load speed silently turns closed-loop control into open loop (2.3).
- **Every hardware change can re-open a closed risk.** The single 3S pack brought back the peak-current problem we had solved in June (5.3).